

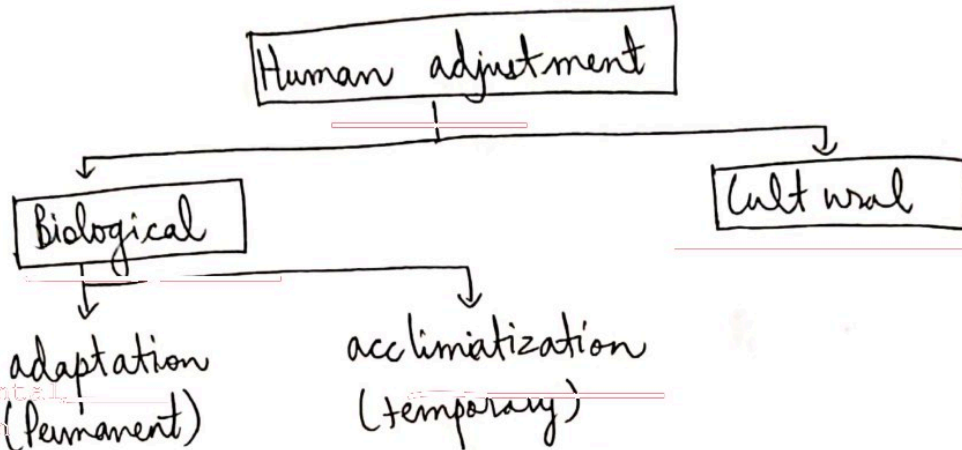
Q.3b (20 marks): Discuss adaptation to environmental stress with examples

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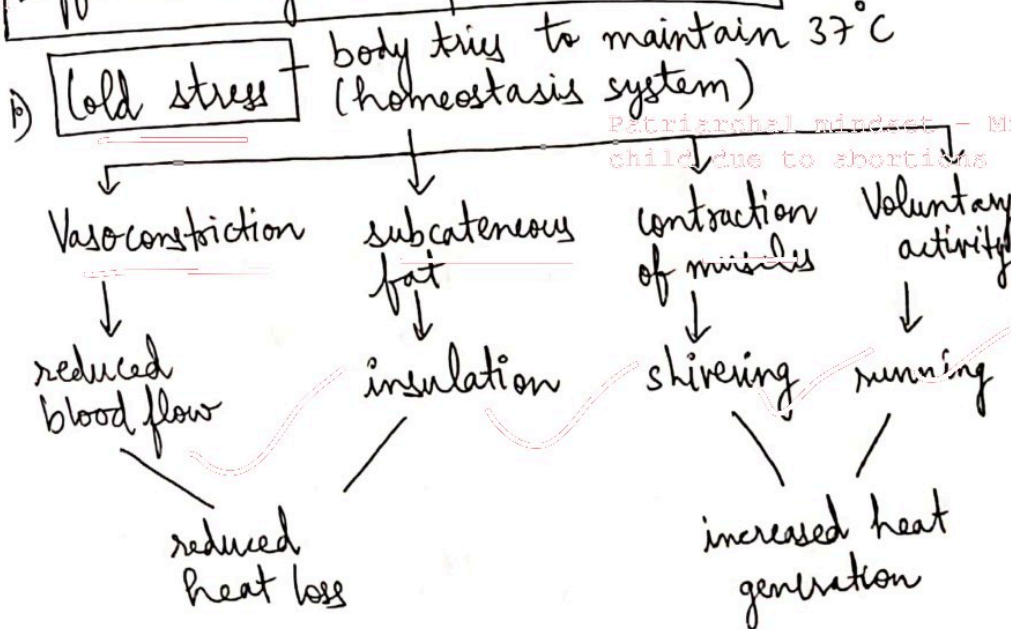
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(b) How do humans adapt? And what are the different ways we have to cope with the stresses of the physical and cultural environments?

Adaptation is the ability to change oneself in order to suit the natural environment.



Different Ways to cope with stress



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Mention about studies on Inuit people

Hunting response : intermittent vasoconstriction and vasodilation protection from frost bite

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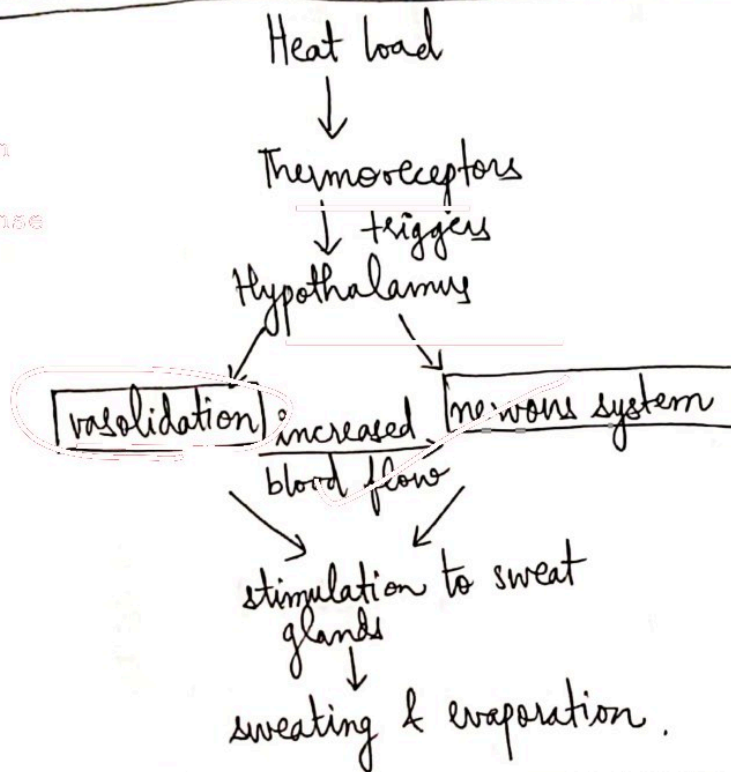
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## 2) Heat stress

Acclimatisation

Sweating response

Tanning



## Environmental adaptation

- 1) Bergmann rule: organisms in warmer areas have relatively smaller size than colder areas.
- 2) Allen rule: organisms in warmer areas have relatively longer appendages than colder areas.

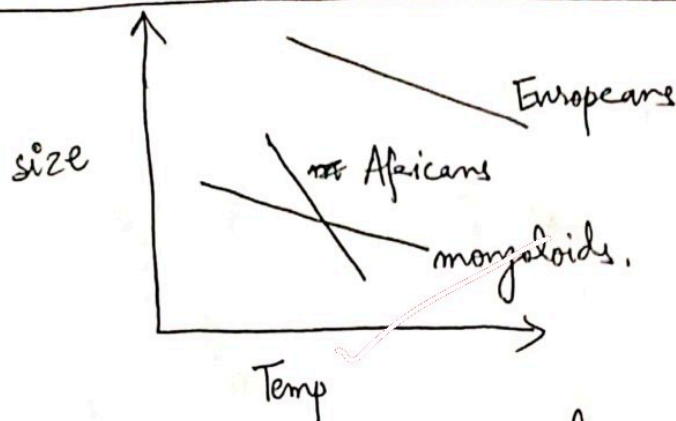


Fig: Allen & Bergman rule

3) Altitude stress: characterized by condition called hypoxia (absence of oxygen)

*Quechua highlanders*

Forced  
Expiratory  
volume

*lowlanders*

Hypoxia

Lungs

higher breathing  
rate

(Little & Mooren  
1990)

Kidneys

Erythropoietin

increased RBC  
& haemoglobin

increased number  
of mitochondria  
& cell energy.

Adaptations

increase in chest size (Monge, 1964)

Decrease in body size

increased ability to respire at  
lower partial pressure

EPAS1 & EGLN1  
gene among Tibetans  
for increased breathing.



You need to mention about adaptation to cultural stress.  
Read the question carefully

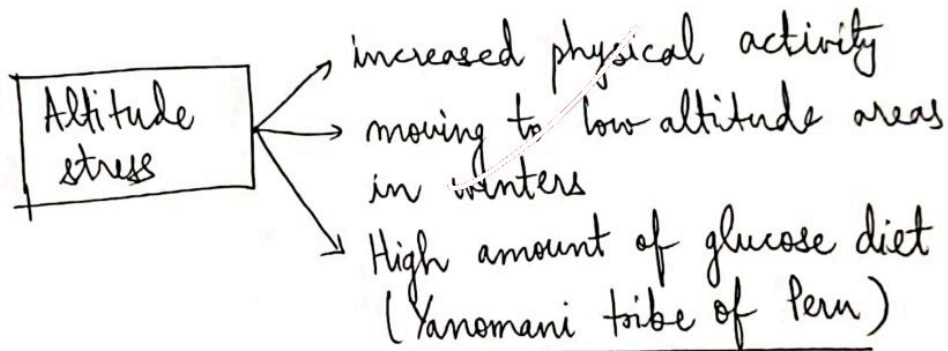
Nutritional deficiency adaptation - Kwashiorkor  
Thrifty genes  
Sickle cell anemia  
(adaptation to malaria - agricultural fields)

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## Cultural adaptation

|                    | <u>Cold stress</u>               | <u>Heat stress</u>  |
|--------------------|----------------------------------|---------------------|
| 1) <u>Clothing</u> | Heavy                            | Light               |
| 2) <u>House</u>    | insulation<br>(Igloos by Inuits) | reflecting material |
| 3) <u>Diet</u>     | High carbohydrate                | High water.         |



Hence, various biological and cultural adaptations are done by humans in order to cope with environmental stresses.